# Program overview

**19-May-2025 10:50**

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| **Year** | **2024/2025** |
| **Organization** | **Civil Engineering and Geosciences** |
| **Education** | **Bridging program Applied Earth Sciences** |

| Code | Omschrijving | ECTS | p1 | p2 | p3 | p4 | p5 |
|---|---|---|---|---|---|---|---|
| **Schakelopleiding Technische Aardwetenschappen 2024** | **Bridging Program Applied Aerth Sciences 2024** | | | | | | |
| **Compulsory courses** | | | | | | | |
| AESB1320-17 | Mechanics | 5 | | | | | |
| AESB2320-24 | Physical Transport Phenomena | 5 | | | | | |
| CT2023 | Programmeren in Python | 3 | | | | | |
| CTB2400 | Numerical Mathematics | 3 | | | | | |
| IFEEMCS010400 | Linear Algebra | 5 | | | | | |
| IFEEMCS010500 | Probability and Statistics | 3 | | | | | |
| IFEEMCS012100 | Calculus for Engineering, part 1 | 3 | | | | | |
| IFEEMCS012200 | Calculus for Engineering, part 2 | 3 | | | | | |
| WI1909TH | Differential Equations | 3 | | | | | |
| **Optional (choose one out of three)** | | | | | | | |
| AESB1130-21 | Geology 1: Basics | 5 | | | | | |
| AESB1230 | Geology 2: North West Europe | 5 | | | | | |
| AESB1440-21 | Methodology of Geophysics and Remote Sensing | 5 | | | | | |

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| **Schakelopleiding Technische Aardwetenschappen 2024** | |
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| **Program Structure 1** | Voor meer informatie, zie ook de regelgeving: http://studenten.tudelft.nl/en/students/faculty-specific/ceg/regulations-ter-rules-and-guidelines/ |

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|---|---|
| **Year** | **2024/2025** |
| **Organization** | **Civil Engineering and Geosciences** |
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**Compulsory courses**

| AESB1320-17 | Mechanics | 5 |
|---|---|---|
| **Responsible Instructor** | Dr. R.E.M. Riva | |
| **Instructor** | Dr.ir. D.S. Draganov | |
| **Instructor** | Dr.ir. B. Wouters | |
| **Contact Hours / Week x/x/x/x** | 0/6/0/0 | |
| **Education Period** | 2 | |
| **Start Education** | 2 | |
| **Exam Period** | 2<br>3 | |
| **Course Language** | English | |
| **Course Contents** | Lectures + analytical exercises.<br>Chapters 2 - 14 Essential University Physics, Volume 1, Richard Wolfson.<br>Subjects: motion in a straight line; motion in two and three dimensions; force and motion; using Newton's laws; work, energy and power; conservation of energy; gravity; system of particles; rotational motion; rotational vectors and angular momentum; static equilibrium; oscillatory motion; wave motion. The acoustic wave equation, stress and strain (hand-outs). | |
| **Study Goals** | Students will be able to understand and to apply basic principles of classical mechanics (including motion in two and three dimensions, forces, Newton's laws, friction, work, energy, conservation of energy, linear momentum, gravity, center of mass, rotational motion, angular momentum, static equilibrium, oscillatory motion, mechanical waves).<br>At the end of this course, students will be able to solve classical mechanics problems (including those about Newton's laws, conservation of energy, linear and angular momentum, gravity, rotational motion, static equilibrium, oscillatory motion, mechanical waves) of difficulty comparable to the difficulty of the problems in the lecture book.<br>In addition, students will be able to describe the acoustic wave equation as used in seismic applications, as well as the concepts of stress and strain. | |
| **Education Method** | A combination of lectures and analytical exercises | |
| **Course Relations** | Fundamental course related to all follow up courses | |
| **Assessment** | Written Exam (100%).<br>For first-year students, admission to the written exam at the end of the lecture period requires completion of the two classroom tests (not graded). | |
| **Remarks** | Lectures: 12x2 =24 hrs<br>Exercises: 8x2=16 hrs<br>Classroom tests: 2x2=4 hours<br>Homework assignments: 6x3=18 hrs<br>Self-study: 75 hrs<br>Exam: 1x3=3 hrs<br>Total: 140 hours | |
| **Expected prior Knowledge** | Mathematics 1 | |
| **Academic Skills** | Critical thinking. Creative use of knowledge to solve new problems. Capacity to work with abstract concepts. Planning and responsibility (timely study crucial to pass exam). | |
| **Literature & Study Materials** | Book: Essential University Physics, Volume 1, by Richard Wolfson, published by Pearson.<br>Hand-outs over the acoustic wave equation. | |
| **Judgement** | 100% written exam (paper based) at the end of the course.<br>For more information on grading, see article 14 in the Rules and Guidelines (RGBE):<br>https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations | |
| **Permitted Materials during Exam** | Open-book exam: use of textbook, hand-outs and lecture slides is allowed.<br>Any other material is not allowed. | |
| **Collegerama** | Yes | |

| AESB2320-24 | Physical Transport Phenomena | 5 |
|---|---|---|
| **Responsible Instructor** | Dr.ir. S.J.A. van der Linden | |
| **Contact Hours / Week x/x/x/x** | 6 hours lectures | |
| **Education Period** | 3 | |
| **Start Education** | 3 | |
| **Exam Period** | 3<br>4 | |
| **Course Language** | English | |
| **Course Contents** | The course Physical Transport Phenomena covers the essential aspects of both fluid flow, and heat and mass transfer at an introductory level, offering you a starting point for your future careers in atmosphere, ocean, and solid earth.<br><br>First, we will look at the basic principles of fluid flow, starting with the different processes that influence the flow and combining them into the conservation equations for mass and momentum. From here, we will treat laminar flow, where attention will be paid to specific examples, including Hagen-Poiseuille flow, Couette flow and Darcys Law. Afterwards, non-dimensional analysis and dimensionless numbers (e.g., Reynolds) are introduced and the topic of turbulence is explored briefly. Bernoulli's Law and the macroscopic energy balance, which are useful for a broad range of engineering applications are treated. Subsequently, attention is focused on heat transfer and the equation for the conservation of energy is derived. Covered subjects include balances for simple heat-transfer problems and different transport mechanisms for heat, for example, heat conduction, convective heat transfer, and radiation.<br><br>During the course, the strong link between education and recent research/realistic practice in engineering will be emphasized. This will be done by giving examples from realistic geophysical flows: hydrological, atmospheric, oceanic, and subsurface applications. | |
| **Study Goals** | The aim of the course is to acquire knowledge about processes of transport of mass, momentum, and energy transfer; and to apply this knowledge to standard problems within the (engineering) field.<br><br>Specifically, after completion of this course, you will be able to:<br>LO 1: describe/recognize different transport mechanisms of mass, momentum and heat.<br>LO 2: describe in a qualitative sense the difference between laminar and turbulent flow, and describe when a flow becomes turbulent.<br>LO 3: analyse a mass, momentum and heat transfer problem, and set up the relevant budget.<br>LO 4: apply the conservation equations of mass, momentum and energy to geophysical applications. | |
| **Education Method** | The course will consist of both lectures (4 hours per week) and guided instruction (2 hours per week). In addition, a number of homework assignments may be given. | |
| **Assessment** | Formative assessment: During the instruction hours, assignments and/or trial exams will be provided that serve as formative assessment for the course.<br><br>Summative assessment: The formal, graded assessment consists of one final assessment worth 5 ECTS (100% of the course). | |
| **Expected prior Knowledge** | - Calculus (AESB1211);<br>- Linear Algebra (AESB1411);<br>- Differential Equations (AESB1210-21);<br>- Mechanics (AESB1320-17);<br>- Electricity & Magnetism (AESB1420-17);<br>- Chemical Thermodynamics (AESB2220-20);<br>- or equivalent. | |
| **Academic Skills** | Calculus, Differential Equations, Vector Analysis, Team-working, Basic Python. | |
| **Literature & Study Materials** | For the course, we use the book: Fundamentals of Momentum, Heat, and Mass Transfer, EMEA Edition, 7th Edition by James Welty, Gregory L. Rorrer, and David G. Foster.<br><br>This can be supplemented by handouts distributed in class and material made available on the Brightspace website. | |
| **Judgement** | Students need to obtain a passing grade for the final written assessment. | |
| **Permitted Materials during Exam** | Students may use a simple calculator during the exam but no computers or communications devices (e.g., phone or tablet computer). Formula sheets, cheat sheets or additional handouts may be provided by the lecturer when necessary. | |
| **Collegerama** | No | |

| CT2023 | Programmeren in Python | 3 |
|---|---|---|
| **Responsible Instructor** | Dr. V.L. Markine | |
| **Contact Hours / Week x/x/x/x** | 2/2/2/2 | |
| **Education Period** | 1<br>2<br>3 | |
| **Start Education** | 1<br>2<br>3 | |
| **Exam Period** | 1<br>2<br>3 | |
| **Course Language** | Dutch | |
| **Collegerama** | No | |

| CTB2400 | Numerical Mathematics | 3 |
|---|---|---|
| **Responsible Instructor** | Prof.dr.ir. C. Vuik | |
| **Co-responsible for assignments** | Dr.ir. D. den Ouden-van der Horst | |
| **Education Period** | 4 | |
| **Start Education** | 4 | |
| **Exam Period** | 4<br>5 | |
| **Course Language** | Dutch | |
| **Expected prior knowledge** | analysis, calculus, linear algebra, differential equations | |
| **Parts** | During the lectures the most important points are highlighted. Motvation for the various parts of the lectures/book are given. Exercises are discussed and explained. There is a lot of material available on the corresponding website for information, motivation, exercises and old exams. | |
| **Summary** | After the course the student is able to design a simple numerical method, analyses it and implement it. Furthermore, the student should be able to consider the results of numerical methods, analyse them and explain their behaviour. An oral and/or written report of these findings can be given by the student. | |
| **Education Method** | Lecture | |
| **Tags** | Mathematics<br>Numeric Methods<br>Programming | |
| **Literature & Study Materials** | Numerical Methods for Ordinary Differential Equations<br>C. Vuik, F.J. Vermolen, M.B. van Gijzen and M.J. Vuik<br>TU Delft OPEN, Delft, 2023<br>https://textbooks.open.tudelft.nl/textbooks/catalog/book/57 | |
| **Judgement** | Grade of the written test is the final grade.<br><br>For more information on grading, see article 14 in the Rules and Guidelines (RGBE):<br>https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations | |
| **Permitted Materials during Exam** | Only a standard calculator. | |
| **Collegerama** | Yes | |

| IFEEMCS010400 | Linear Algebra | 5 |
| --- | --- | --- |
| **Responsible Instructor** | Dr. E. Emsiz | |
| **Contact Hours / Week x/x/x/x** | 6/x/x/x | |
| **Education Period** | 1 | |
| **Start Education** | 1 | |
| **Exam Period** | 1<br>2 | |
| **Course Language** | Dutch<br>English | |
| **Expected prior knowledge** | See the requirements of the specific MSc programma students are applying for. | |
| **Summary** | Course email: ifeemcs010400@tudelft.nl<br>(please don't use personal email addresses) | |
| **Course Contents** | - Solving systems of linear equations (using Gaussian elimination)<br>- Vectors in R^n<br>- Linear dependence/independence<br>- Matrix algebra (sum, product, inverse)<br>- LU decomposition of a matrix<br>- Linear transformations<br>- Subspaces in R^n (basis, dimension); column space and null space<br>- Determinants<br>- Eigenvalues and eigenvectors<br>- Diagonalization<br>- Inner products and orthogonal sets<br>- Orthogonal projections<br>- Method of least squares and linear models<br>- Symmetric matrices | |
| **Study Goals** | An important goal of the course is to make the student comfortable with fundamental notions from linear algebra. At the end of this course you will be able to understand the connections between these concepts. You may also be asked to come up with proofs for elementary properties (i.e. write down a consistent/correct short proof) and give a correct counterexample.<br><br>After successfully completing the course you will be able to:<br><br>- Solve systems of linear equations.<br>- Identify if a system of linear equations is solvable, and if the solution is unique.<br>- Show the equivalence between a vector equation, a matrix equation and a system of linear equations.<br>- Give a geometric description of solutions sets of vector equations.<br>- Determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence.<br>- Identify if a transformation is linear and determine the standard matrix of a linear transformation.<br>- Determine the matrix of geometric linear transformations.<br>- Be able to perform matrix operations (sum, scalar multiple, multiplication, transpose) and describe the law-like properties of matrix multiplication and apply these properties.<br>- Describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics.<br>- Give definitions of subspace, column space and null space of a matrix and determine a basis for a subspace.<br>- Describe and apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem.<br>- Determine the LU-decomposition of a matrix.<br>- Determine the determinant of a matrix by applying the properties of a determinant.<br>- Apply the properties of determinants (product, transpose, inverse).<br>- Determine the eigenvalues and eigenvectors of a matrix.<br>- Determine the algebraic and geometric multiplicity of a matrix.<br>- Determine if a matrix is diagonalizable and if so, determine a diagonalization of the matrix.<br>- Analyze the case of complex eigenvalues and eigenvectors.<br>- Determine the matrix of a linear transformation with respect to a basis.<br>- Calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets.<br>- Calculate the orthogonal projection on a subspace.<br>- Determine an orthogonal basis of a subspace by means of the Gram-Schmidt process.<br>- Explain the normal equation of the least-squares method and apply the least-squares method to linear models.<br>- Explain and calculate the orthogonal diagonalization of symmetric matrices and describe the spectral theorem for symmetric matrices. | |
| **Education Method** | Colstruction | |
| **Literature and Study Materials** | Book: David C. Lay et. al., Linear Algebra and its Applications, Global Edition, 6/E.<br>- Slides on Brightspace.<br>- Prelecture videos.<br>- Online homework assignments.<br>- Book exercises<br>- Old exams on Brightspace.<br><br>Disclaimer: information may change depending on the developments around the coronavirus. | |
| **Assessment** | Final score: F<br><br>IF the midterm score M is higher than the score E for the exam AND E >= 5.0<br>THEN F = 0.25 M + 0.75 E<br>ELSE F = E<br><br>All grades are rounded to 1 decimal place.<br><br>Note that the midterm is not mandatory (and cannot be resited).<br><br>For the resit the midterm is not relevant anymore. | |
| **Permitted Materials during Tests** | None | |
| **Test and result scale** | 1-10 | |
| **Co-Instructor** | Dr.ir. R.F. Swarttouw | |

| IFEEMCS010500 | Probability and Statistics | 3 |
|---|---|---|
| **Responsible Instructor** | Dr. R.J. Fokkink | |
| **Responsible Instructor** | Dr. C. Kraaikamp | |
| **Contact Hours / Week x/x/x/x** | x/x/4/x or x/x/x/4 | |
| **Education Period** | 3<br>4 | |
| **Start Education** | 3<br>4 | |
| **Exam Period** | 3<br>4<br>5 | |
| **Course Language** | Dutch<br>English | |
| **Expected prior knowledge** | Linear Algebra, Calculus for Engineering part 1 and part 2 or Calculus for Science part 1 and part 2. | |
| **Course Contents** | Graphical and numerical summaries of data<br>Introduction to probability<br>Some elementary combinatorics to find probabilities<br>Conditional probability; Bayes' rule; stochastic (in)dependence<br>Random variables; probability mass function; density function; distribution function<br>Standard distributions: Binomial, Poisson, Geometric, Normal, Uniform, Exponential, Pareto<br>Expectation and variance; transformations; Jensen's inequality<br>Multivariate random variables; joint distributions; joint and marginal density functions; (in)dependence of random variables<br>Covariance and correlation<br>Chebychev's inequality; Law of Large numbers; Central Limit Theorem<br>Sampling theory and statistical models; mean, sample variance, histogram, empirical distribution function, boxplot<br>Theory of Estimators: Bias, Efficiency, Mean squared error<br>Linear Regression: univariate and multivariate, goodness of fit<br>Introduction to statistical learning<br>Confidence intervals of Mean and Proportion; t-distribution<br>Testing theory: Type I/II Error, p-value, significance level, critical region<br>One sample t-test | |
| **Study Goals** | At the end of the course, you will be able to:<br><br>- compute (conditional) probabilities<br>- use discrete and continuous random variables<br>- use joint random variables<br>- compute expectation and (co)variance<br>- apply the law of large numbers and the central limit theorem<br>- analyse data by means of graphical and numerical summaries<br>- formulate a statistical model<br>- estimate parameters<br>- perform a linear regression analysis<br>- construct a confidence interval<br>- formulate and carry out hypothesis tests<br>- Use a computer program to perform elementary simulation and data analysis | |
| **Education Method** | Interactive lectures, pre-lecture videos, exercises | |
| **Literature and Study Materials** | Book: Mathematical introduction to Probability and Statistics, Dekking et al. | |
| **Assessment** | One midterm (week 5) on Part 1: Probability Theory and one endterm (week 10) on Part 2: Statistics. Final grade is the average of both tests. To pass the course, the grade for part 2 needs to be at least 4 or higher. | |
| **Permitted Materials during Tests** | 1) Formula sheet to be used for exams of Probability and Statistics<br>2) Any kind of calculator | |
| **Remarks** | Please note that this course is available in Q3 and in Q4. | |
| **Co-Instructor** | Dr. C. Kraaikamp | |
| **Co-Instructor** | Dr. R.J. Fokkink | |

| IFEEMCS012100 | Calculus for Engineering, part 1 | 3 |
|---|---|---|
| **Responsible Instructor** | Dr. E. Emsiz | |
| **Contact Hours / Week x/x/x/x** | 4/x/x/x | |
| **Education Period** | 1 | |
| **Start Education** | 1 | |
| **Exam Period** | 1<br>2 | |
| **Course Language** | Dutch<br>English | |
| **Expected prior knowledge** | See the requirements of the specific MSc programma students are applying for. | |
| **Co-Instructor** | Dr.ir. N.P.K. Chan | |

| IFEEMCS012200 | Calculus for Engineering, part 2 | 3 |
|---|---|---|
| **Responsible Instructor** | Dr. E. Emsiz | |
| **Contact Hours / Week x/x/x/x** | x/4/x/x | |
| **Education Period** | 2 | |
| **Start Education** | 2 | |
| **Exam Period** | 2<br>3 | |
| **Course Language** | Dutch<br>English | |
| **Co-Instructor** | Dr.ir. N.P.K. Chan | |

| WI1909TH | Differential Equations | 3 |
|---|---|---|
| **Responsible Instructor** | Drs. I.A.M. Goddijn | |
| **Contact Hours / Week x/x/x/x** | 0/4/0/0 | |
| **Education Period** | 2 | |
| **Start Education** | 2 | |
| **Exam Period** | 2<br>3 | |
| **Course Language** | Dutch | |
| **Expected prior knowledge** | Calculus and Linear Algebra | |
| **Course Contents** | Linear differential Equations, Systems of first order, linear and not-linear differential equations, Partial differential equations and Fourier series. | |
| **Study Goals** | Learning a number of mathematical techniques necessary for solving various technical-physical model equations. | |
| **Education Method** | Colstruction | |
| **Books** | Elementairy Differential Equations and Boundary Value Problems,<br>Global Edition<br>ISBN-13: 978-1-119-38287-4<br>(preferable)<br><br>or<br><br>Elementairy Differential Equations and Boundary Value Problems,<br>International Adaptation<br>ISBN-13: 978-1-119-82051-2 | |
| **Assessment** | Exam with open questions. Due to circumstances (such as Covid-19) this can become an online exam, even in a different format. In case of insufficient results a repair option may exist in accordance with Article 2, Examination requirements, Clause 4, of the Implementation Regulations 2023-2024. | |
| **Permitted Materials during Tests** | Announced during lectures and on Brightspace | |
| **Co-Instructor** | Dr.ir. Z.J.G. Gromotka | |

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|---|---|
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| **Organization** | **Civil Engineering and Geosciences** |
| **Education** | **Bridging program Applied Earth Sciences** |

**Optional (choose one out of three)**

| AESB1130-21 | Geology 1: Basics | 5 |
|---|---|---|
| **Responsible Instructor** | Dr. G. Bertotti | |
| **Instructor** | Drs. J.C. Blom | |
| **Contact Hours / Week x/x/x/x** | 7/0/0/0 | |
| **Education Period** | 1 | |
| **Start Education** | 1 | |
| **Exam Period** | 1<br>2 | |
| **Course Language** | English | |
| **Parts** | The Geology 1 course is composed of two components dedicated to 1) general knowledge of the system Earth, 2) tools for the 3D geometric representation of geological objects. | |
| **Course Contents** | General knowledge of the system Earth, tools for the 3D geometric representation of geological objects.<br><br>Part 1: the rock cycle, the structure and functioning of the Earth and its outer layer, the lithosphere; horizontal deformations and vertical movements; the climate system; notions of hydrogeology; sedimentary systems from the source to the sink (marine systems).(Bertotti)<br>Part 2: focuses on tools to represent the architecture of geologic bodies on maps and how these can be used to infer the geological evolution of a specific region (Blom) | |
| **Study Goals** | The goal of the Geology I course is to provide students basic knowledge and skills to understand relevant geological processes and enable them to make geological predictions required by their future careers.<br><br>Knowledge:<br>Students will be able to:<br>- Describe the first order structure and functioning of the Earth. This includes basic knowledge of i) elements, minerals and rocks, ii) of the internal structure of the Earth with particular attention to the lithosphere and the crust, and iii) of plate tectonics.<br>- Describe the main characteristics of sedimentary, igneous and metamorphic rocks and will have an understanding of how these rocks were formed.<br>- Describe kinematics and basic physics of vertical movements (subsidence creating sedimentary basins and uplift creating relief) and horizontal deformations at lithospheric to thin section scale. They will be able to quantify simple relations between relief and crustal structure. They will be able to extract deformation scenarios from simple geological documents such as geological and seismic sections.<br>- Describe exogene processes and their impact on the sedimentary record. Processes include climate, terrestrial water flow and oceanography.<br>- Describe the main sedimentary environments and their evolution through time. Students will also be able to infer such evolutionary schemes from simple geological records such as stratigraphic columns and use them for simple paleogeographic reconstructions.<br><br>Skills. Students will be able to:<br><br>- Interpret geologic maps, geologic sections and other basic documents such as seismic section to infer the 3D architecture of the area/volume of investigation<br>- Use the techniques mentioned above to constrain larger scale processes and make predictions for spatially and temporally unknown parts of the Earth system (mainly Earth upper crust). | |
| **Education Method** | The course Geology 1 is a combination of lectures and laboratory work with practicals with maps, physical samples etc.<br>In total there will be 28hrs of lectures, 35hrs of practicals, 4,5hrs of tests and 3hrs for the repetition. | |
| **Assessment** | The final grade results from two components: 1) General geology, the Basics (60%) and 2) Geological sections and maps (40%).<br>The grade of component 1 is composed of the results of the practical and of the exam. The percentages will be defined at the beginning of the course<br>The grade of component 2 (Geological maps and sections) is given on the basis of the exam.<br><br>A grade of 5 or more required for each of the two components.<br><br>Independent resits for the two components will be held | |
| **Permitted Materials during Tests** | To be specified in the lectures before tests. | |
| **Contact** | Prof. Dr. G. Bertotti<br>G.bertotti@tudelft.nl<br>tel: +31 (0)15 27 86033 | |
| **Expected prior Knowledge** | none - introductory course | |
| **Academic Skills** | understanding and reading English texts; use basic mathematics tools and physical concepts | |
| **Literature & Study Materials** | The book adopted for the Geology 1 course is Understanding Earth (Grotzinger and Jordan)<br>Handouts of lectures will be available on Brightspace | |
| **Judgement** | For more information on grading, see article 14 in the Rules and Guidelines (RGBE):<br>https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations | |
| **Permitted Materials during Exam** | Coloured pencils, rulers and scales | |
| **Collegerama** | No | |

| AESB1230 | Geology 2: North West Europe | 5 |
|---|---|---|
| **Responsible Instructor** | Drs. J.C. Blom | |
| **Contact Hours / Week x/x/x/x** | Q3: 4 hours lecture & 4 hours practical | |
| **Education Period** | 2 | |
| **Start Education** | 2 | |
| **Exam Period** | 2<br>3 | |
| **Course Language** | English | |
| **Expected prior knowledge** | Geology I | |
| **Course Contents** | Minerals and Rocks, Geology of NW Europe, Structural geology. | |
| **Study Goals** | The goal of the Geology 2 course is to provide students with knowledge and skills to understand the processes over time that lead to the formation of North West Europe, as well as the identification of some minerals and rocks that can be found in outcrop and in the subsurface of North West Europe and elsewhere | |
| **Study Goals continuation** | Knowledge:<br>Students will be able to:<br>- Explain the history and structure of the subsurface of North West Europe, as well as the natural processes that led to the formation of the present day North West Europe, with special emphasis on the Netherlands.<br>- Describe the origin, formation and location of the different types of geologic resources found in North West Europe.<br>- Explain the processes that lead to the formation of rocks (igneous, metamorphic and sedimentary).<br>- Describe rock characteristics (texture, mineralogy) and use this to decide on rock names.<br>- Demonstrate connections between rock forming processes and rocks (rock cycle).<br>- Describe the tectonic processes and different deformational styles that played a role in the formation North West Europe, as well as other regions.<br>- Explain the processes that play a role in the deformation of the Earth's crust.<br><br>Skills:<br>Students will be able to:<br>- macroscopically identify 20 different rock forming minerals<br>- Analyze the texture and mineralogy of rocks (igneous, metamorphic and sedimentary) macroscopically and subsequently able to infer the way they formed and to identify the rocks.<br>- Draw cross sections illustrating the subsurface of North West Europe | |
| **Education Method** | The course Geology 2 is a combination of self study, lectures, laboratory work with practical experience on the macroscopic rock determination, a practical with maps and report writing.<br><br>The Geology 2 course is composed of three parts dedicated to:<br>1) general knowledge and practical skills needed for the recognition of different minerals and rock types;<br>2) general knowledge of the Geology of North West Europe;<br>3) general knowledge and practical skills in Structural Geology.<br>These last two parts will be taught in an integrated fashion, with a combination of self study, the writing of reports, and feedback during the contact hours. | |
| **Literature and Study Materials** | Part 1 will use powerpoints and instructions via Brightspace.<br>For parts 2 and 3 powerpoints of lectures will be made available on Brightspace. In addition material from "Geology of the Netherlands", which will be available through Brightspace, and emodules from "Structural Geology", by Haakon Fossen (Cambridge University Press) will be used during class. | |
| **Books** | none mandatory, although Structural Geology by Haakon Fossen may be helpful. | |
| **Reader** | all information will be made available on Brightspace in the form of powerpoints and links to websites. | |
| **Prerequisites** | Geology 1 | |
| **Assessment** | All three parts of the course will be tested with a written exam.<br><br>Judgement: All three blocks count equally for the final grade.<br><br>--- UPDATE dd 2-2-2021 due to the COVID-19 measures ---<br>The course will be assessed with three online tests. | |
| **Permitted Materials during Tests** | None | |
| **Remarks** | The General Geology is valid for 5 EC which corresponds to 140 hrs, that is, 53 contact hours and 87 hrs of self-study.<br>The 53 hours will be organized as follows:<br>36 hrs of lectures/feedback sessions (pt.1:12; pt.2: 12; pt.3: 12)<br>12 hrs of practical work (pt. 1: 12;)<br>5 hours: Three tests will be held at the end of each block<br><br>The self-study hours will be divided as follows:<br>About 21 hours of preparation for intermediate tests (7 per test), about 66 hours for preparation/working out lecture hours and practicals. | |
| **Contact** | Drs. J.C Blom<br>J.C.Blom@tudelft.nl<br>tel: +31 (0)15 27 83628 | |
| **Expected prior Knowledge** | AESB 1130, Geology 1 | |
| **Academic Skills** | For the Structural and NWE part, students will be asked to prepare before the scheduled contact hours, so these hours can be used to ask questions regarding the parts that were not understood.<br>In addition to this, students will be asked to perform a weekly research project on the geology of part of NW Europe, and present the results of this research in a report, on which they will receive feedback. | |
| **Literature & Study Materials** | | |
| **Judgement** | All three tests count equally for the final grade.<br><br>For more information on grading, see article 14 in the Rules and Guidelines (RGBE):<br>https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations | |
| **Permitted Materials during Exam** | none | |

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| **Collegerama** | No |

| AESB1440-21 | Methodology of Geophysics and Remote Sensing | 5 |
|---|---|---|
| **Responsible Instructor** | Dr.ir. G.G. Drijkoningen | |
| **Instructor** | Dr. P.G. Ditmar | |
| **Instructor** | Dr.ir. G.G. Drijkoningen | |
| **Instructor** | Dr. J.M. Duran | |
| **Instructor** | W.B. Vermaat | |
| **Contact Hours / Week x/x/x/x** | Q4: approx. 9 hours per week<br><br>Applied Geophysics & Remote Sensing Methodology: 20 hours<br>Integrated project: 20 hours<br>Philosophy of Science lectures by TBM: 5 hours | |
| **Education Period** | 4 | |
| **Start Education** | 4 | |
| **Exam Period** | 4<br>5 | |
| **Course Language** | English | |
| **Course Contents** | Students are given a broad overview of Applied-Geophysics and Remote-Sensing methods. In addition, the general, scientific basis underlying such methodology is introduced (e.g., the interplay between data, models and theory). The course includes separate lectures on philosophy of science such that students can critically assess the scientific method. The methods are studied in a societal and/or economically relevant context, and some of them are applied in a field experiment.<br><br>In groups, the students will work on three large assignments (problem-based learning):<br><br>1. An in-depth study of a specific applied-geophysics method (e.g., refraction/reflection seismics, electric resistivity imaging, etc.) in the context of a societally and/or economically relevant problem. The groups are expected to find resources themselves, which will train them to find and value applied-geophysics literature, and independently acquire relevant knowledge. This assignment commences with a general introduction (lecture) in applied-geophysics methodologies. To monitor the group dynamics, ensure progress and focus, and provide feedback, several consults are scheduled over the course of the assignment. A draf report is submitted which is also provided with feedback, enabling students to shape their writing skills. At the end a final report is submitted and the results are presented orally.<br><br>2. An in-depth study of a specific remote-sensing method (e.g., gravimetry, hyperspectral imaging, etc.) in the context of a societally and/or economically relevant problem. The groups are expected to find resources themselves, which will train them to find and value remote-sensing literature, and independently acquire relevant knowledge. This assignment commences with a general introduction (lecture) in remote-sensing methodologies. To monitor the group dynamics, ensure progress and focus, and provide feedback, several consults are scheduled over the course of the assignment. A draf report is submitted which is also provided with feedback, enabling students to shape their writing skills. At the end a final report is submitted and the results are presented orally.<br><br>3. Field application of some of the applied-geophysics and remote-sensing methods studied in assignments 1 and 2. For applied geophysics, often the electric resitivity method, low-frequency EM or (shear-wave) seismic method is taken, while precise positioning with GPS/GNSS will be taken as the Remote-Sensing method. This assignment includes some literature study (bibliographical study), defining goals and hypotheses, modelling of the experiments, planning of the field experiments, carrying out the field experiments, processing of the field data, and interpretation of the data and results. The results are subsequently compared against the initial hypothesis (modelling results), and conclusions are drawn in terms of the initial goals (geological/environmental in nature). The entire project (workflow) and its results are presented orally (after the lecturers have given feedback on the draft presentation).<br><br>Importantly, the oral presentations at the end of the assignments will familiarise all students with the different applied-geophysics and remote-sensing methods (and hence not only with the methodology they studied with their group). | |
| **Study Goals** | Knowledge: Students will be able to<br>- apply fundamental knowledge (mathematics, physics, geology) in some elementary Applied-Geophysics and Remote-Sensing projects.<br>- relate basic physical laws (on mechanics and electricity & magnetism) to applied-geophysics and remote-sensing methods.<br>- classify different Applied-Geophysics and Remote-Sensing methods.<br>- link theoretical concepts, resulting from basic physical laws, with practices in the field.<br>- use an applied-geophysics and remote-sensing method to address a societal, scientifically, or economically relevant problem.<br><br>Skills: students will be able to<br>- simulate the result of field experiments using forward modelling techniques.<br>- make a field-acquisition plan, acquire, process and interpret data obtained with an applied-geophysical and remote-sensing method.<br><br>Soft skills: students will be able to:<br>- approach a problem from a scientific point of view, using basic physical laws, observations/facts and models (philosophy of science).<br>- write a scientific report.<br>- give an oral presentation of a scientific subject.<br>- retrieve and use relevant literature from the library for its use in a geoscience project.<br><br>Attitude: students will be able to<br>- work efficiently in groups not of ones own choice; to work together and delegate tasks; to be accountable to their group; and to deal with limitations of fellow group members. | |
| **Education Method** | Problem-based approach. Students will work in groups of 3 to 4. They will give oral presentations, carry out library/literature studies and write reports. They will apply knowledge/skills from philosophy of science to problems/projects.<br><br>There will be a few introductory lectures about applied-geophysics and remote-sensing methodologies to support the problem-based education. The separate consults with the groups, and the (formative) feedback on the drafts of the presentations and reports, allow the students to meet the learning objectives (or study goals). | |
| **Course Relations** | This course has a relation to the courses:<br>- Geophysical Methods for Subsurface Characterization (2nd year)<br>- Instrumentation & Signals (2nd year)<br>- Geological Fieldwork & Remote Sensing (2nd year)<br>- Field Exploration & Exploitation (3rd year)<br>- Applied-Geophysics and Remote-Sensing courses in the MSc program | |
| **Assessment** | - Applied-geophysics and remote-sensing methods (assignments 1 & 2): written report (both assignments) plus an oral presentation by the group (to all fellow students) of one of the two assignments. | |

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| | - Integrated field project (assignment 3): oral presentation (to all fellow students) by the group.<br><br>- Separate student-individual summative assessment is performed by a lecturer of TBM (including formative feedback). Consequently, each student in a group has to give at least one oral presentation.<br><br>- Philosophy of Science is assessed separately.<br><br>The final grade is then built up as follows:<br>- Applied-geophysics and remote-sensing methodology (assignments 1 & 2): 60%  (Register for AESB1440-21 Exam 01)<br>- Integrated field project (assignment 3): 30% (Register for AESB1440-21 Exam 02)<br>- Presentation skills: 5% (Register for AESB1440-21 Exam 03)<br>- Philosophy of Science: 5% (Register for AESB1440-21 Exam 04) |
| **Special Information** | The total study load is 5 EC. This is divided as follows (see also assessment):<br>Applied-geophysics and remote-sensing methodology: 3.0 EC  (Exam 01)<br>Integrated project (including self-acquired field data): 1.5 EC  (Exam 02)<br>Presentation skills (soft skills): 0.25 EC  (Exam 03)<br>Philosophy of Science: 0.25 EC  (Exam 04) |
| **Contact** | Guy Drijkoningen<br>G.G.Drijkoningen@tudelft.nl |
| **Expected prior Knowledge** | Since this is a course in which a large number of geophysical and remote sensing methods are introduced, almost all courses in the first year of the AES pogramme are relevant. The following, however, are particularly important:<br>- Grand challenges and applied Earth sciences (for the report writing and presentation skills)<br>- Mathematics 1<br>- Mechanics<br>- Electricity and magnetism |
| **Academic Skills** | Scientific skills: approach a problem from a scientific point of view (Philosophy of Science)<br>Soft skills: report writing (assessed), oral presentation (assessed) and library skills (feedback) |
| **Literature & Study Materials** | Acquiring relevant knowledge is part of the learning objectives. Of course, the students will be supported by the lecturers for this. When necessary, relevant articles and information will be made available via Brightspace. |
| **Judgement** | For more information on grading, see article 14 in the Rules and Guidelines (RGBE):<br>https://www.tudelft.nl/en/student/ceg-student-portal/education/education-information/educational-rules-and-regulations |
| **Permitted Materials during Exam** | (no exam: only reports and presentations) |
| **Collegerama** | No |

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